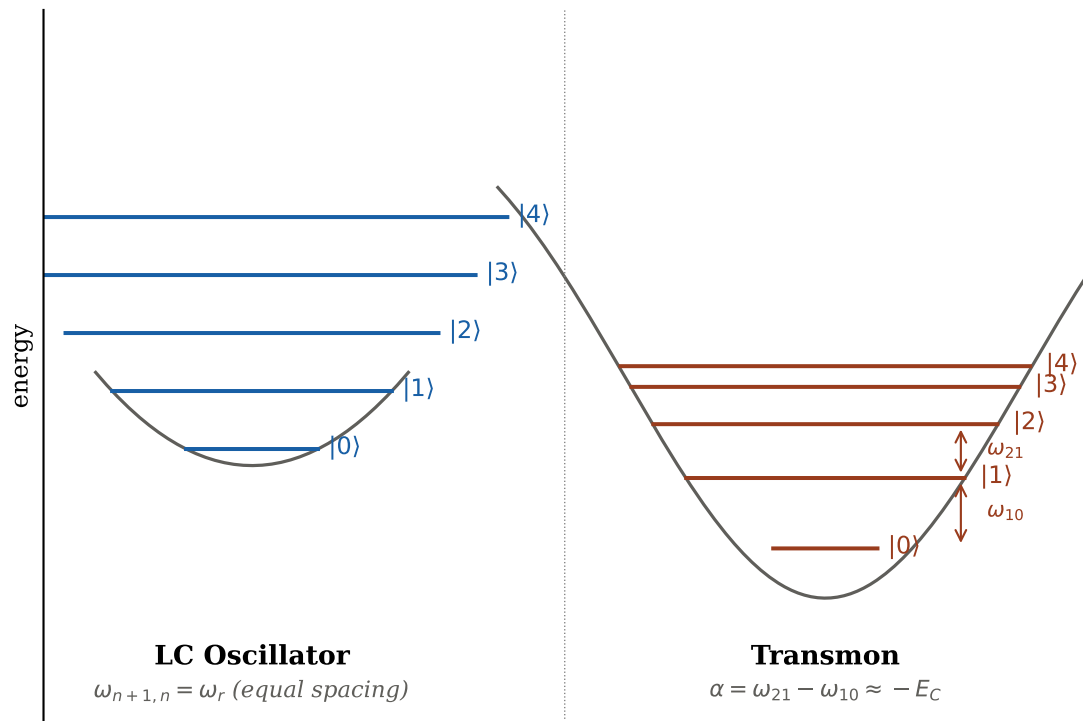
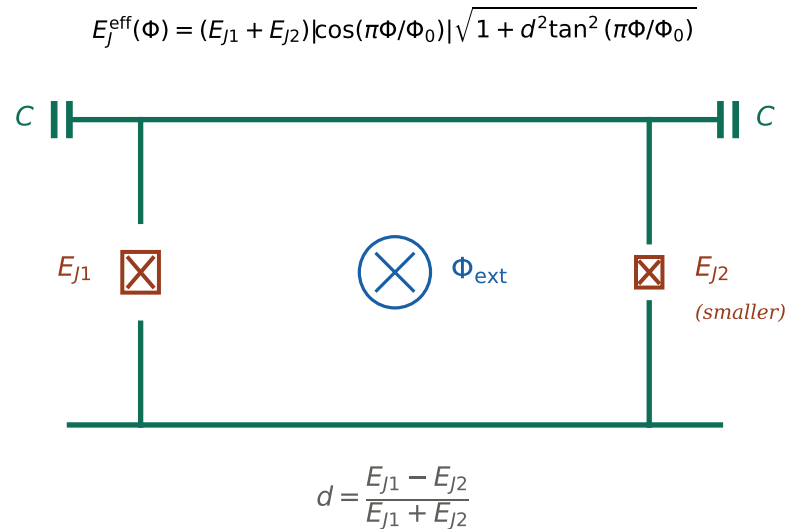


(a) anharmonicity makes the transmon a qubit



(b) asymmetric SQUID with external flux Φ_{ext}



two flux-insensitive sweet spots per period if $d \neq 0$:
upper at $\Phi = 0$, lower at $\Phi = \Phi_0/2$ with $E_J^{\text{eff}} = (E_{J1} + E_{J2})|d|$